Perl Modules Used *for* Web Designing

This is the second article in the 'The Resurgence of Perl' series, in continuation from the November issue of *OSFY*. Currently, there's a lot of focus on the cloud and the Web environment designed for it. This article highlights two former modules of Perl for Web designing, and how successful companies still relish this old approach!

In the last article, we had already set up Apache to run a script named *printenv*, using *mod_cgi* and *mod_perl*. Let us compare these modules in a little more detail for the same script. Those who have not read the first article in this series can find it at *https://opensourceforu.com/2018/11/the-resurgence-of-perl/*.

As per my experience, these modules are the best CGI wrappers for Perl, when used with Apache (as compared to any other Web servers).

If you are using a virtual box or any other method of hosting your own GNU/Linux server, apart from AWS or any cloud service, we will march forward assuming that the required scripts in the last article were all run without errors, and that Apache is running with *mod_perl* also configured in it. This is what enables us to run the same script from the same folder but a different URL alias, and thereby view the differences.

```
printenv
#!/usr/local/bin/perl
print "Content-type: text/plain; charset=iso-8859-1\n\n";
for each \$var (sort(keys(%ENV))) {
 print "\$var=".\$ENV{\$var}."\n";
}
```

In fact, we can have one more script in the same folder as the above, with the same ownership and permissions. Name the script *printenv-html*.

```
#!/usr/local/bin/perl

print "Content-type: text/html\n\n";
foreach \$var (sort(keys(%ENV))) {
 print "\$var=".\$ENV{\$var}."<br>";
}
```

Let's now compare *printenv* and *printenv-html*.

a) *print "Content-type: text/plain; charset=iso-8859-1\n\n";* became *print "Content-type: text/html\n\n";*

i. STDOUT of Perl converts to HTML.

ii. The font size and type changes because it has become HTML (the default HTML font size is 1cm or 16px, and non-HTML font size is 11px).

Note: The default content type for Internet mail is *"text/plain; charset=us-ascii"*. This is the reason why automated emails sent from any programming language, when rendered by a GUI based mail client, do not use the default font of the mail client and do not look that appealing. Also, we cannot control the look and feel of the email if we do not change it to text/HTML. How to send HTML mail using Perl has been covered in an earlier article, the reference for which is given at the end of the article.

b) *print "\$var=".\$ENV{\$var}."\n";* became *print "\$var=".\$ENV{\$var}."
";*

Perl STDOUT (text/plain) understands that '\n' is supposed to represent a new line but text/HTML does not. In HTML, the 'br' tag means a new line. If we use '\n', then the output will be in one line.

Now we have four different cases:

1. *http://IPAddress/cgi-bin/printenv*
2. *http://IPAddress/cgi-bin/printenv-html*
3. *http://IPAddress/cgi-perl/printenv*
4. *http://IPAddress/cgi-perl/printenv-html*

In *printenv* and *printenv-html*, the difference is only in how the output looks and feels. For this article, we will consider cases 1 and 3.

Table 1 shows the various environment variables that *printenv* prints. The number of variables might increase based on whether it was authenticated with a specific